INCH-POUND
MIL-DTL-0060944B (AR)
05 April 2004
USED IN LIEU OF
MIL-C-60944A (AR)
15 SEPTEMBER 1980

DETAIL SPECIFICATION

CARTRIDGE, 40MM - PRACTICE, M385A1

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-C-60944A (AR).

1. SCOPE

1.1 Scope. This specification covers the metal parts and loading, assembling and packaging for one type of cartridge designated as Cartridge, 40MM: Practice, M385A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-60942	- Primer, Percussion for 40MM
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys
MIL-A-70625	- Automated Acceptance Inspection Equipment Design, Testing and Approval of

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

U.S. ARMY (ARDEC) DRAWINGS (see 6.6)

PRODUCT AND PACKAGING DRAWINGS

8886326 - Cartridge, 40MM: Practice, M385A1
8886405 - Linking of 40 MM High Velocity Ammunition
12928042 - Packing and Marking of Shipping and Storage Container PA120 with Linked 40MM Cartridge

INSPECTION EQUIPMENT DRAWINGS

9202253 - Chamber Gage
9202255 - Gage, Plug, Acceptance
9202254 - Gage, Plug, Wear Limit Check
9202929 - Action Time

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM B117 - Standard Method of Salt Spray (Fog)
ASTM E1255 - Standard Practice for Radioscopy
ASTM D3951 - Standard Practice for Commercial Packaging

(Copies of this document are available online at www.astm.org or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Air pressure. The cartridge shall withstand an air pressure of three (3) pounds-per-square-inch (p.s.i.) for five (5) seconds minimum (min.) without a drop in pressure.

3.4 Functioning. The cartridge assembly shall function, not cause any weapon stoppages due to the operation of loading, uncoupling, firing and ejecting, and comply with the following:

3.4.1 Velocity. When fired from the MK19, MOD III, Grenade Machine Gun, the mean velocity of the sample cartridges shall be 790 ± 10 feet-per-second (fps) and a standard deviation not exceeding 12 fps.

3.4.2 Security of crimp of base plug and action time. When fired from the M129 Grenade Launcher, the action time of the cartridge shall not exceed 4 milliseconds and there shall be no base plug movement more than .005 above flush with the rear of cartridge case after firing.

3.5 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.6 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement/Inspection Cross Reference Matrix

Method of inspection
N/A - Not Applicable
1 - Analysis
2 - Demonstration
3 - Examination
4 - Test

Classes of inspection
A - First Article
B - Conformance

Section 3 Requirement	Inspection Methods					Inspection Class		Section 4 Inspection Method
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4, 4.5.1, 4.5.2, 4.5.3, 4.5.5, 4.5.7
3.3					X	X	X	4.5.6
3.4.1					X	X	X	4.5.8
3.4.2				X	X	X	X	4.5.4
3.5				X		X	X	4.5.10
3.6				X		X	X	4.5.11

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article. When specified in the solicitation or contract, a sample of cartridges, 40MM: practice, M385A1 and subcomponents shall be subjected to first article inspection in accordance with Table II and 4.3.1.

4.3.1 First article quantity. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provision of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawings	Quantity
Cartridge Case Assembly	8886327	25
Cartridge	8886326	240
Linked Ammunition	8886405	50

4.3.2 Inspections to be performed. The first article inspection shall be 100 % examination of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table II herein.

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4.3.3 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

TABLE II. First Article Inspection

40MM Cartridge, Practice M385A1			
EXAMINATION OR TEST	N0. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Cartridge (Dwg. 8886326)			
Air Pressure test	50	3.3	4.5.6
Pull Test	50	3.2	4.5.7
X-ray examination of ballistic sample	240	3.2	4.5.5
Functioning	240		4.5.8
- Single Shot	144	3.4	4.5.8.1
- Rapid Fire	96	3.4	4.5.8.2
Notes:			

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with the requirements of MIL-STD-1916. For the components identified below, each inspection lot of cartridges shall be produced with components complying with the following constraints:

a. Projectile metal parts from one lot interfix number from one manufacturer
b. M169 cartridge case loading assemblies metal parts from one lot interfix from one manufacturer
c. Primers from one lot number from one manufacturer
d. Propellant from one lot number from one manufacturer
e. Links from one lot interfix number and one manufacturer
f. Only cartridges from one lot per ammunition belt shall be used. No short belts permitted

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.8. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE Cartridge Case Assembly, Prior to Loading	SHEET 1 OF 1		DRAWING NUMBER 8886327
				NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Closing cup missing	100%	3.2	AAIE / 4.5.9
2.	Three (3) or more vent holes blocked or Missing 2/	100%	3.2	AAIE / 4.5.9
MAJOR				
101.	One (1) or two (2) vent holes blocked or missing 2/	100%	3.2	AAIE / 4.5.9
102.	Two or more closing cups 1/	100%	3.2	AAIE / Radiography / 4.5.9
MINOR				
201.	Evidence of Poor Workmanship	Level II	3.6	Visual / 4.5.11

NOTES:
1/ May be inspected at higher level assembly.
2/ Acceptance inspection of the cartridge case is with AAIE by the case manufacturer. Verification at the LAP facility is required.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.2	TITLE Cartridge Case Assembly Prior to Assembling Base Plug and Primer	SHEET 1 OF 1		DRAWING NUMBER 8886327
				NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL 1.	Propellant charge weight less than 75 percent or more than 125 percent of assessed weight 1/	100%	3.2	AAIE / 4.5.9 or 4.5.2
MAJOR 101.	Propellant charge weight out of drawing tolerance but within ± 25 percent of assessed weight 1/	100%	3.2	AAIE / 4.5.9 or 4.5.2 / 4.5.3
MINOR 201.	Evidence of Poor Workmanship	Level II	3.6	Visual / 4.5.11

NOTES:
1/ May be inspected at lower level of assembly provided Government approval is obtained.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.3	TITLE Projectile Assembly, prior to assembly to Cartridge Case Assembly	SHEET 1 OF 1		DRAWING NUMBER 8866714
				NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR 101.	O-ring missing from groove	Level IV	3.2	Visual
MINOR 201.	Evidence of Poor Workmanship	Level II	3.6	Visual / 4.5.11
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.4	TITLE Cartridge Case Assembly	SHEET 1 OF 1		DRAWING NUMBER 8886327
				NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Primer above flush with base plug	100%	3.2	AAIE / 4.5.9
MAJOR				
101.	Base plug above flush	Level IV	3.2	Gage
102.	Depth of base plug, max	Level IV	3.2	Gage
103.	Depth to primer, max., from base plug	Level IV	3.2	Gage
104.	Excessive sealing compound on exterior surface of primer	Level IV	3.2	Visual
105.	Security of crimp of base plug and action time 2/	80 3/	3.2	4.5.4 / 9202929
106.	Primer damaged	Level IV	3.2	Visual with min. 3X magnification
107.	Check test for deterioration of primers	As required 1/	3.2	4.5.1
MINOR				
201.	Evidence of poor workmanship	Level II	3.6	Visual / 4.5.11

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-P-60942.
2/ May be completed prior to assembling projectile assembly.
3/ Accept on zero (0) and reject on one (1).

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.5	TITLE Cartridge, 40MM, M385A1	SHEET 1 OF 1		DRAWING NUMBER 8886326
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	X-ray examination of ballistic sample 1/	240 Acc 0 Rej 1	3.2	4.5.5
MAJOR				
101.	Chamber gage failure	100%	3.2	Gage / 9202253 9202254 9202255
102.	Total Length, maximum	100%	3.2	Gage
103.	X-ray examination of ballistic sample 2/	240 3/	3.2	4.5.5
104.	Air pressure test	Level III	3.2	4.5.6
105.	Pull test of projectile	Level III	3.2	4.5.7
106.	Gap, maximum, at rear of rotating band	Level IV	3.2	Gage
107.	Packing exposed	Level IV	3.2	Visual
108.	Functioning, MK19, MOD III, machine gun (Single Fire)	144 4/	3.4	4.5.8.1
109.	Functioning, MK19, MOD III, machine gun (Rapid Fire)	96 4/	3.4	4.5.8.2
MINOR				
201.	Marking missing, improper or illegible	Level II	3.2	Visual
202.	Evidence of Poor Workmanship	Level II	3.6	Visual / 4.5.11

NOTES:
1/ A critical defect is defined as a low or high propellant charge or a missing closing cup.
2/ A major defect is defined as any defect not specified in Note 1/.
3/ Any major defect found shall be reported with the exception of two (2) closing cups which shall result in the rejection of the lot and ballistic sample.
4/ See Table III for classification and acceptance / rejection criteria.

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.6	TITLE Linked Ammunition, Prior to Packing	SHEET 1 OF 1		DRAWING NUMBER 8886405
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR 101.	Linked improperly 1/	100%	3.2	Visual
MINOR 201.	Evidence of poor workmanship	Level II	3.6	Visual / 4.5.11

NOTES:
1/ Anyone of the four (4) tabs of link not in groove; link inverted; link damaged; link distorted.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.7	TITLE Shipping and Storage Container (unsealed)		SHEET 1 OF 1	DRAWING NUMBER 12928042
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Cartridge inverted	Level IV	3.2	Visual
102.	Incorrect number of cartridges	Level IV	3.2	Visual
103.	Fillers missing	Level IV	3.2	Visual
104.	Support, top and bottom missing (when applicable)	Level IV	3.2	Visual
105.	Spacer, missing	Level IV	3.2	Visual
106.	Top pad missing (when applicable)	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.6	Visual / 4.5.11
NOTES:				

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Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.8	TITLE Shipping and Storage Container (sealed)		SHEET 1 OF 1	DRAWING NUMBER 12928042
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Box or container damaged	Level IV	3.2	Visual
MINOR				
201.	Contents loose	Level II	3.2	Manual
202.	Can seal missing or improperly positioned	Level II	3.2	Visual
203.	Marking incorrect or illegible	Level II	3.2	Visual
204.	Evidence of poor workmanship	Level II	3.6	Visual / 4.5.11
NOTES:				

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TABLE III. Function Defect Classification and Acceptance/Rejection Criteria

	Acceptance Number		Rejection Number		Classification of Defect
	MK19 Single Shot 3/	MK19 Rapid Fire	MK19 Single Shot 3/	MK19 Rapid Fire	
1. Failure to meet Mean Velocity	1/	--------	2/	--------	Major
2. Standard deviation of velocity over max.	1/	--------	2/	-------	Major
3. Primer leak as evidence by perforation of primer cup or smoke deposit on base plug or cartridge case	0	0	1	1	Major
4. Projectile lodges in weapon bore	0	0	1	1	Critical
5. Primer misfire	0	0	1	1	Major
6. Weapon stoppage excluding primer misfire	--------	0	--------	1	Major

Notes:

1/ Lot acceptable if requirement is met.
2/ Lot rejected if requirement is not met.
3/ M129 Grenade Launcher is used to measure Action Time & Security of crimp of base plug.

4.5 Methods of inspections.

4.5.1 Check test for deterioration of primers. If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-60942 immediately before the primer lot is assembled into the cartridge. The test shall be performed at the facility assembling the primers into the cartridge.

4.5.2 Propellant weight. The propellant quantity shall be determined and then weighed 100 percent. Any charge that fails to comply with the drawing requirement (taking into account machine accuracy) shall be classified defective and removed from the lot. The accuracy of the machine shall be determined and the prove-out test performed to validate the machine's ability to remove charges outside of the drawing tolerance range. The machine must be capable of detecting and rejecting critical propellant weights, such that the end product contains less than 1 in a million critical defects (examples machine error rate not to exceed 1/500 and maximum defect rate not to exceed 0.05 percent).

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4.5.3 Propellant weight alternate method. At the start of production, 380 consecutive samples per station, per machine, shall be selected and check weighed 100 percent. All samples shall comply with the applicable drawing requirement. If all samples comply with the applicable drawing requirement, 5 samples per station, for every 20 minutes production, shall be selected and the weight of each of the 5 samples must meet the applicable drawing requirement. If any weight fails the applicable drawing requirement, the 20 minutes production represented by that sample shall be rejected and set aside for disposition in accordance with MIL-STD-1916. Correction will be made and 380 consecutive samples check weighed from that station. If all samples meet the applicable requirement, sampling every 20 minutes shall be resumed.

4.5.4 Security of crimp of base plug and action time. The cartridge assemblies shall be assembled to a projectile and fired from a M129 Grenade Launcher. After firing observation shall be made for base plug movement and action time. If any base plug movement or action time exceeds the applicable requirement, or any projectile lodges in the bore, the quantity represented shall be rejected and set aside for disposition in accordance with MIL-STD-1916. The movement of the base plug shall be measured from the highest point on the base plug to a point on the cartridge case directly beside it. The cartridge case assemblies produced by each primer assembly machine shall be kept segregated and identified. They shall not be used in production until successful completion of this test.

4.5.5 X-ray examination of ballistic sample. Prior to the forwarding of the cartridges to a government approved facility for ballistic testing, the cartridges shall be inspected for defects using approved x-ray equipment. This test shall be conducted from the side. If any critical defect is found, the lot shall be rejected including the ballistic sample and set aside for disposition in accordance with MIL-STD-1916. This test shall be conducted in accordance with ASTM 1255.

4.5.6 Air pressure. The cartridge assembly shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observations shall be made for a drop in pressure indicating a leak. This is a non-destructive test. Parts successfully tested may be returned to the lot.

4.5.7 Pull test of projectile. The cartridge shall be placed in an approved fixture and the axial tension specified on the applicable drawing shall be applied. If the sample fails the drawing requirement, the lot represent by the sample shall be rejected and set aside for disposition in accordance with MIL-STD-1916, root cause assemblies may be reused after undergoing a Government approved rework provided the same crimping method is used. Projectile assemblies if undamaged may be reused. The cartridge shall be pulled until separation occurs and the data observed.

WARNING: This test is conducted on assemblies containing an explosive element. Test barricades (where used), procedures and equipment shall have prior Government approval.

4.5.8 Function test. The test shall be conducted at a government approved installation. If new, the MK19, MOD III, Grenade Machine Gun and barrel shall have at least forty (40) rounds fired on it prior to being used for the test. The distance to impact medium of coarse sand shall be 200 meters plus or minus (±) 5 meters.

4.5.8.1 Single shot. The barrel selected for the test must be used throughout the entire test. The rounds shall be conditioned at 70 ± 5º F for at least (8) hours and then fired within 15 minutes of being removed from conditioning temperature. The cartridges shall be fired single shot from a MK 19, Mod III, Grenade Machine Gun at a rate not to exceed ten (10) per minute. At least six (6) gun warmers shall be rapid fired at the beginning of the test. The velocity shall be measured at 30 feet

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from the muzzle of the launcher. If an approved radar system is not used, then the monitoring devices and time recorders shall be a dual system setup for recording two (2) separate readings that shall be within 3 feet per second of each other. If the two readings differ by more than 3 feet per second (for a given shot), the results shall be declared a "no test" and another round fired in its place. The number of "no test" rounds shall be reported. "No test" rounds shall be reported as "no test" rounds for velocity readings. Report the approximate range to impact of "no test" rounds. Results shall be calculated and recorded for mean velocity and standard deviation (see 6.7). Observation shall be made for compliance with Table III.

4.5.8.2 Rapid Fire. The cartridge shall be rapid fired from a MK19, MOD III, Grenade Machine Gun mounted on the MK64 Mount and M3 Tripods. Firing shall be in burst of five (5) to ten (10) rounds. Rate of fire shall be recorded for informational purpose only. Observation shall be made for compliance with Table III.

4.5.9 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out shall be performed in accordance with MIL-A-70625 (see 6.8).

4.5.10 Ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL-STD-1168.

4.5.11 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.6.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the metal parts and loading, assembling and packaging for one type of cartridge designated as Cartridge, 40MM: Practice, M385A1.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number (w/revision level) and date of all reference documents.

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c. ASSIST Online to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see Section 2).
d. Requirements for submission of first article sample.
e. Requirement for submission of design verification sample, if applicable.
f. Requirement for submission of inspection equipment designs.
g. Applicable national stock number.
h. Packaging requirements, if other than specified in Section 5.
i. Serialization requirements, if applicable.
j. Certificate of conformance for each lot or shipment of product, if applicable.
k. Government-Furnished Material or equipment, if applicable.
l. Provisions for critical characteristics controls.
m. Critical Inspection Equipment Requirement (See 6.8)

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. Inspection equipment, inspection method and the frequency of inspection should be submitted for approval to ARDEC, ATTN: AMSRD-QEM-D, Picatinny, NJ 07806-5000.

6.4 Contractor acceptance inspection equipment (AIE). Provision concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract.

6.5 Ammunition lot numbers. Ammunition requires ammunition data cards in accordance with MIL-STD-1168.

6.6 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Standard deviation can be calculated from the following formula:

$$s = \sqrt{\Sigma (x-x^{bar})^2/n-1}$$

Where:

x = each individual value

x^{bar} = sample arithmetic mean = x/n

$\Sigma (x-x^{bar})^2$ = The sum of squares of the differences between the sample mean and each individual value.

n-1 = each individual value

n = sample

6.7.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used.

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$s = \sqrt{n \Sigma x^2 - (\Sigma x)^2 / n(n-1)}$

Other approved standard deviation formulas may be used.

6.8 Critical Inspection Equipment.

6.8.1 System Reliability. The contractor should provide for each defect identified as critical, a manufacturing and inspection system that assures no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.8.2 Maximum defect rate produced. The maximum defect rate produced should be defined as the largest defect rate expected for the characteristic of concern on a monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and re-inspected.

6.8.3 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The contractor, to meet the system reliability as defined in 6.8.1, defines the maximum error rate allowed. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop a demonstration test procedure to demonstrate the error rate of the equipment. The test should be performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the required error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.)

6.8.4 Periodic verification. Periodic verification of the system error rate should be performed.

6.8.5 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.8.6 Automated acceptance inspection equipment. Equipment for which no, or minimal, human involvement is required in the acceptance determination. This would include equipment employing probes/sensors/transducers that are automatically manipulated to perform measuring and detection functions. If readout is provided, the equipment will be categorized as automated even if operator interpretation is needed in the accept/reject decision.

6.9 Definitions.

6.9.1 Reliability. The number of successes (rounds chambered, fired, and ejected) divided by the number tested. Occasionally the last round of a continuous belt, due to the light weight of the cartridge case, will not fully eject from the weapon. This type of failure should not be classified as a

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weapon stoppage nor should this type of failure be recorded as a defect chargeable to the ammunition.

6.10 Subject term (Key word) listing.

Target Practice (TP)
MK19, MOD III, Grenade Machine Gun
M129 Grenade Machine Gun

6.11 Submission of alternative conformance provisions. All contractor proposed alternative conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.12 Revisions /Amendments History. The following is a record of changes incorporated into this document

ECP	Description	Date Approved
M6Q2031	1. Provides the means for a first article testing 2. Provides changes to the requirements 3. Provides changes to quality levels because of the transition from MIL-STD-105 to MIL-STD-1916 4. Provides additional verification methods 5. Provides changes to format	08/19/1996
R9S3055	Delete requirement for dual velocity measuring system when using Doppler radar	01/13/2000
R04Q2014	Initial Release of M385A1 Cartridge MIL-DTL Specification	Aug 2004

6.13 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Army-AR
(Project 1310-2006-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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